SET A

Aim:Write a program to implement TextView and EditText.

Q1:

Write and execute code to display user input from EditText into TextView.

ANS :

Activity_main.xml

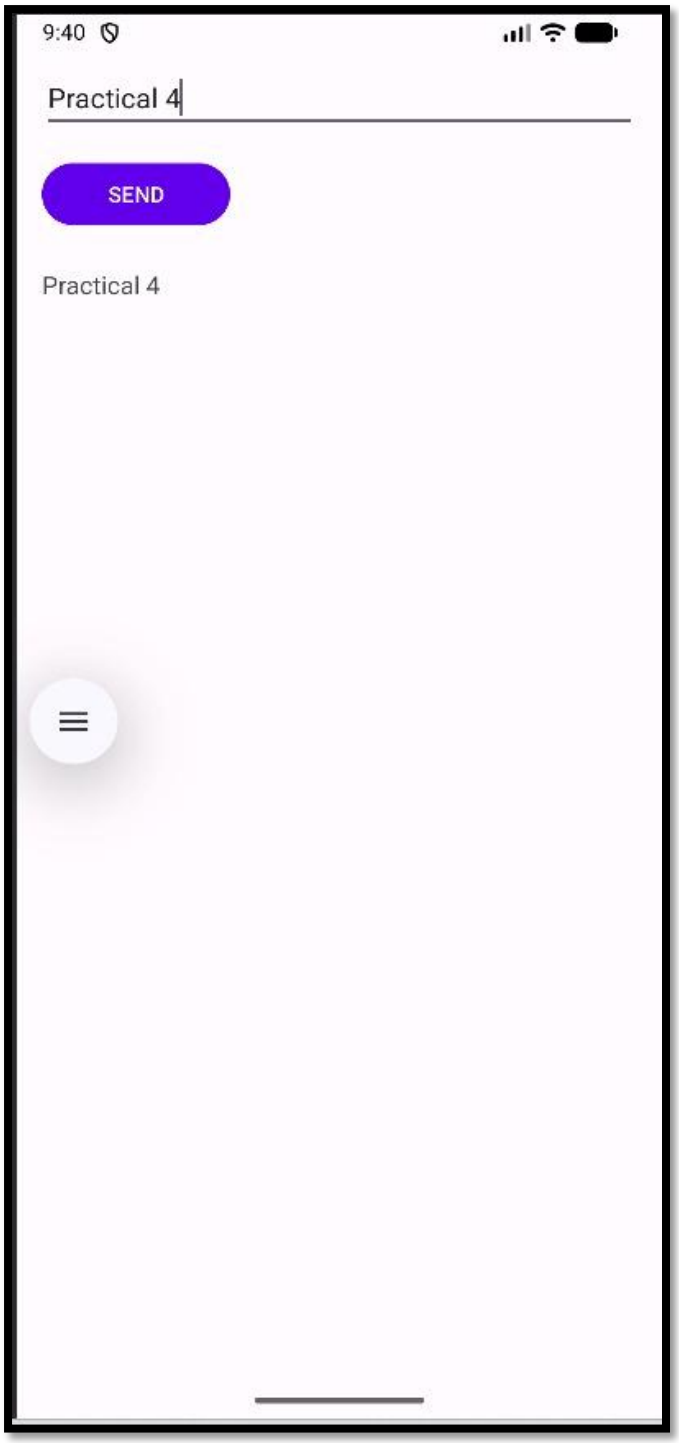
```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:padding="16dp">

  <EditText
    android:id="@+id/editTextInput"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter text"
    android:layout_marginTop="16dp"
    android:layout_marginBottom="16dp" />

  <Button
    android:id="@+id/buttonDisplay"
    android:layout_width="120dp"
    android:layout_height="wrap_content"
    android:text="SEND"
    android:backgroundTint="#6200EE"
    android:textColor="@android:color/white" />
```

```
  <TextView
    android:id="@+id/textViewOutput"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:layout_marginTop="24dp"
    android:textSize="16sp"
    android:visibility="gone" />

</LinearLayout>
```

MainActivity.java

```
package com.example.myapplication;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

  EditText etInput;
  TextView tvOutput;
  Button btnDisplay;

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
```

```
        etInput = findViewById(R.id.editTextInput);
        tvOutput = findViewById(R.id.textViewOutput);
        btnDisplay = findViewById(R.id.buttonDisplay);

        btnDisplay.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View view) {
                String text = etInput.getText().toString();
                tvOutput.setText(text);
                tvOutput.setVisibility(View.VISIBLE);
            }
        });
    }
}
```



Q2:

Differentiate between TextView and EditText and explain their important attributes.

- **TextView is used to display text, while EditText is used to take input from the user.**
- **TextView is non-editable, whereas EditText is editable.**

- **TextView is used for labels and messages, while EditText is used in forms and input fields.**
- **Important attributes of TextView: *text, textColor, textSize, gravity*.**
- **Important attributes of EditText: *hint, inputType, text, maxLength*.**

SET B

Aim: Write a program to implement Button, ImageButton and ToggleButton.

Q1:
Create and execute a layout with all three buttons: Button, ImageButton and ToggleButton.

ANS :

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:padding="16dp"
  android:gravity="center">

  <Button
    android:id="@+id/button"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Button"
    android:layout_marginBottom="16dp" />
```

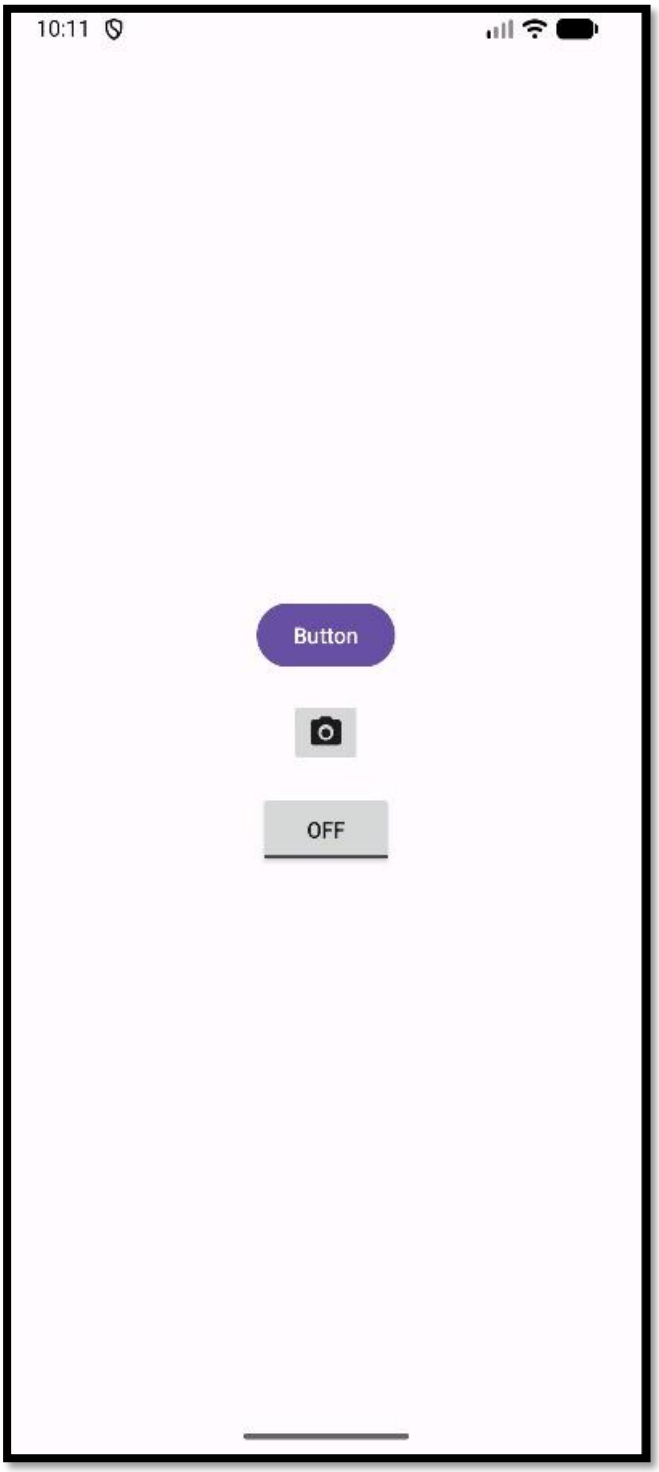
```
    <ImageButton
        android:id="@+id/imageButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:src="@drawable/ic_button_image"
        android:contentDescription="Image Button"
        android:layout_marginBottom="16dp" />

    <ToggleButton
        android:id="@+id/toggleButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:textOff="OFF"
        android:textOn="ON" />

</LinearLayout>
```

MainActivity.java

```
package com.example.myapplication;

import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```
        setContentView(R.layout.activity_main);
    }
}
```

Output:



Q2:
What are Button, ImageButton, and ToggleButton in Android? Explain their differences.

ANS

- **Button is a UI element used to perform an action when clicked. It usually displays text.**
- **ImageButton is similar to Button but displays an image instead of text and is used for graphical actions.**
- **ToggleButton is a button with two states (ON/OFF) and is used to switch between states.**

Button, ImageButton, and ToggleButton (Difference Table)

Feature	Button	ImageButton	ToggleButton
Definition	Used to perform an action with text	Button with an image instead of text	Button with two states (ON/OFF)
Display	Text	Image	Text (ON/OFF)
Function	Single click action	Single click action	Switch between two states
State	No state	No state	Has ON/OFF state
Usage	Submit, click actions	Icons, graphical UI	Enable/disable features

Important Attributes

- **Button → *text, onClick, background***
- **ImageButton → *src, contentDescription, background***
- **ToggleButton → *textOn, textOff, checked*.**

SET C

Aim: Write a program to implement CheckBox.

Q1:
Write and execute code to display selected hobbies using CheckBox.

ANS :

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <CheckBox
      android:id="@+id/check1"
      android:text="Reading"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content" />

    <CheckBox
      android:id="@+id/check2"
      android:text="Gaming"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content" />

    <Button
      android:id="@+id/btn"
      android:text="Show"
      android:layout_width="wrap_content"
      android:layout_height="wrap_content" />

    <TextView
      android:id="@+id/result"
      android:layout_width="match_parent"
      android:layout_height="wrap_content" />

</LinearLayout>
```

MainActivity.java

```
package com.example.myapplication;


import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.CheckBox;
import android.widget.TextView;


public class MainActivity extends AppCompatActivity {


  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);


    CheckBox check1 = findViewById(R.id.check1);
    CheckBox check2 = findViewById(R.id.check2);
    TextView result = findViewById(R.id.result);


    findViewById(R.id.btn).setOnClickListener(v -> {
      String text = "";
      if (check1.isChecked()) text += "Reading ";
      if (check2.isChecked()) text += "Gaming ";
      result.setText(text.isEmpty() ? "None" : text);
    });
  }
}
```

Output:

Q2:

Explain CheckBox and its usage in Android.

ANS

- **CheckBox in Android**
- **CheckBox is a UI component that allows the user to select one or more options from a set.**
- **It works independently, meaning multiple CheckBoxes can be selected at the same time.**
- **It is commonly used for multiple-choice selections.**

- Usage of CheckBox
- Used in forms to select hobbies, interests, or preferences.
- Allows users to choose multiple options (e.g., Reading, Music, Sports).
- Helps in gathering user input where more than one selection is allowed.
- It can be checked or unchecked based on user interaction using methods like isChecked().

- Important Attributes
- text → Displays label of CheckBox
- checked → Sets default state (true/false)
- id → Unique identifier for referencing in code

SET D

Aim: Write a program to implement ProgressBar.

Q1:
Design and execute an Android app that displays a horizontal progress bar updating every second until it reaches 100%.

ANS :

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:gravity="center"
  android:padding="20dp">

  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
```

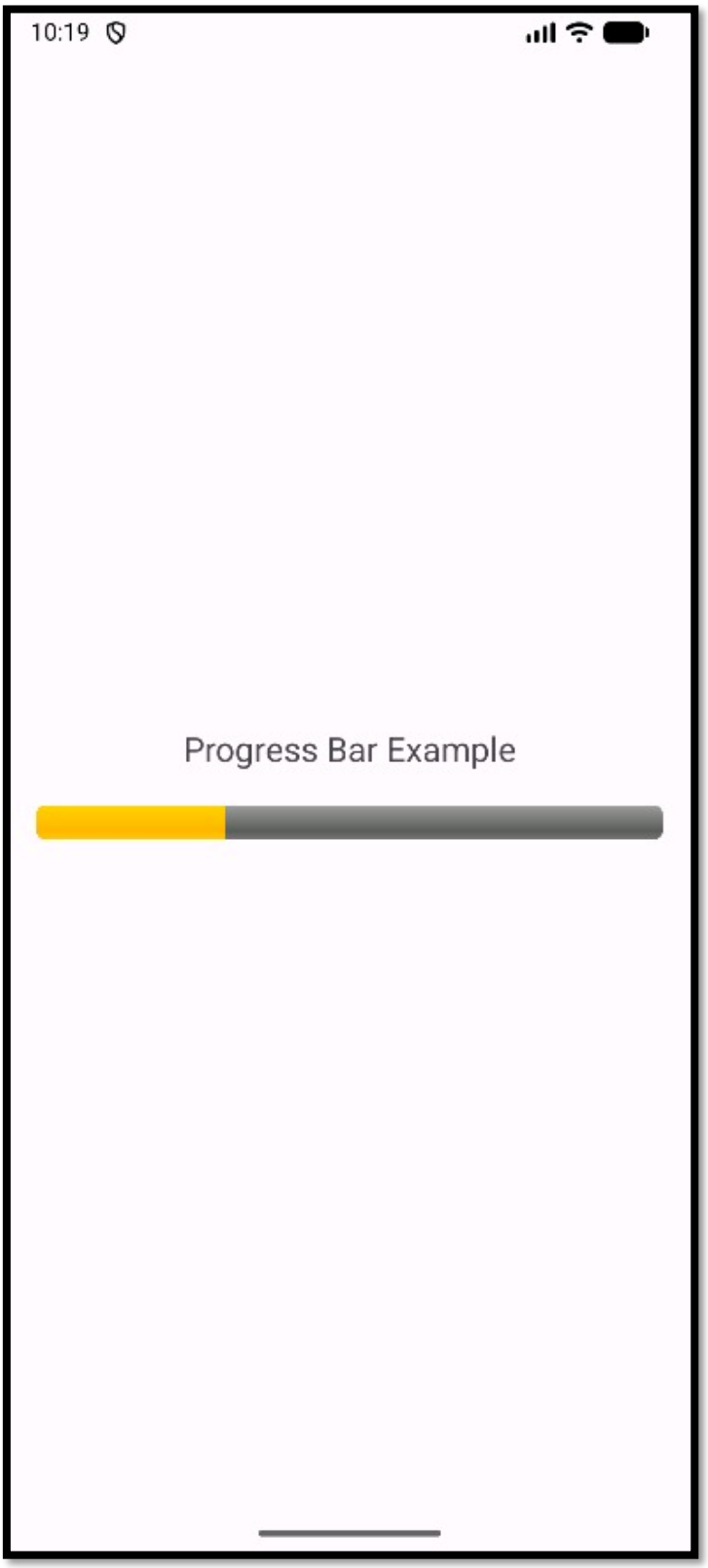
```
    android:text="Progress Bar Example"
    android:textSize="20sp"
    android:layout_marginBottom="20dp" />

  <ProgressBar
    android:id="@+id/progressBar"
    style="@android:style/Widget.ProgressBar.Horizontal"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:max="100"
    android:progress="0" />

</LinearLayout>
```

Main_Activity.java

```
package com.example.myapplication;

import android.os.Bundle;
import android.os.Handler;
import android.widget.ProgressBar;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

  ProgressBar progressBar;
  int progress = 0;
  Handler handler = new Handler();
```

```
@Override
protected void onCreate(Bundle savedInstanceState) {
  super.onCreate(savedInstanceState);
  setContentView(R.layout.activity_main);

  progressBar = findViewById(R.id.progressBar);

  new Thread(new Runnable() {
    @Override
    public void run() {
      while (progress < 100) {
        progress += 10;
        handler.post(new Runnable() {
          @Override
          public void run() {
            progressBar.setProgress(progress);
          }
        });
        try {
          Thread.sleep(1000);
        } catch (InterruptedException e) {
          e.printStackTrace();
        }
      }
    }
  }).start();
}
```

Output:-



Q2:
Explain ProgressBar in Android and list its types.

ProgressBar in Android

- **ProgressBar is a UI component used to show the progress of an ongoing task.**
- **It helps users understand that a process is running (like loading, downloading, etc.).**
- **It can be displayed in horizontal or circular form.**

Types of ProgressBar

- **Determinate ProgressBar**
  - **Shows exact progress (e.g., 50%, 70%).**
  - **Used when the task duration is known.**
- **Indeterminate ProgressBar**
  - **Does not show exact progress.**
  - **Displays continuous animation (loading spinner).**
  - **Used when task duration is unknown.**

Important Attributes

- **max → Maximum value of progress**
- **progress → Current progress value**
- **indeterminate → Defines whether progress is known or not**

SET E

Aim: Write a program to implement a converter application.

Q1:
Create and execute a temperature converter from Celsius to Fahrenheit.
ANS :

activity_main.xml

```xml
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
  android:layout_width="match_parent"
  android:layout_height="match_parent"
  android:orientation="vertical"
  android:padding="20dp">

  <TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Temperature Converter"
    android:textSize="20sp"
    android:textStyle="bold" />

  <EditText
    android:id="@+id/etCelsius"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Enter temperature in Celsius"
    android:inputType="numberDecimal" />

  <Button
    android:id="@+id/btnConvert"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Convert to Fahrenheit" />

  <TextView
```

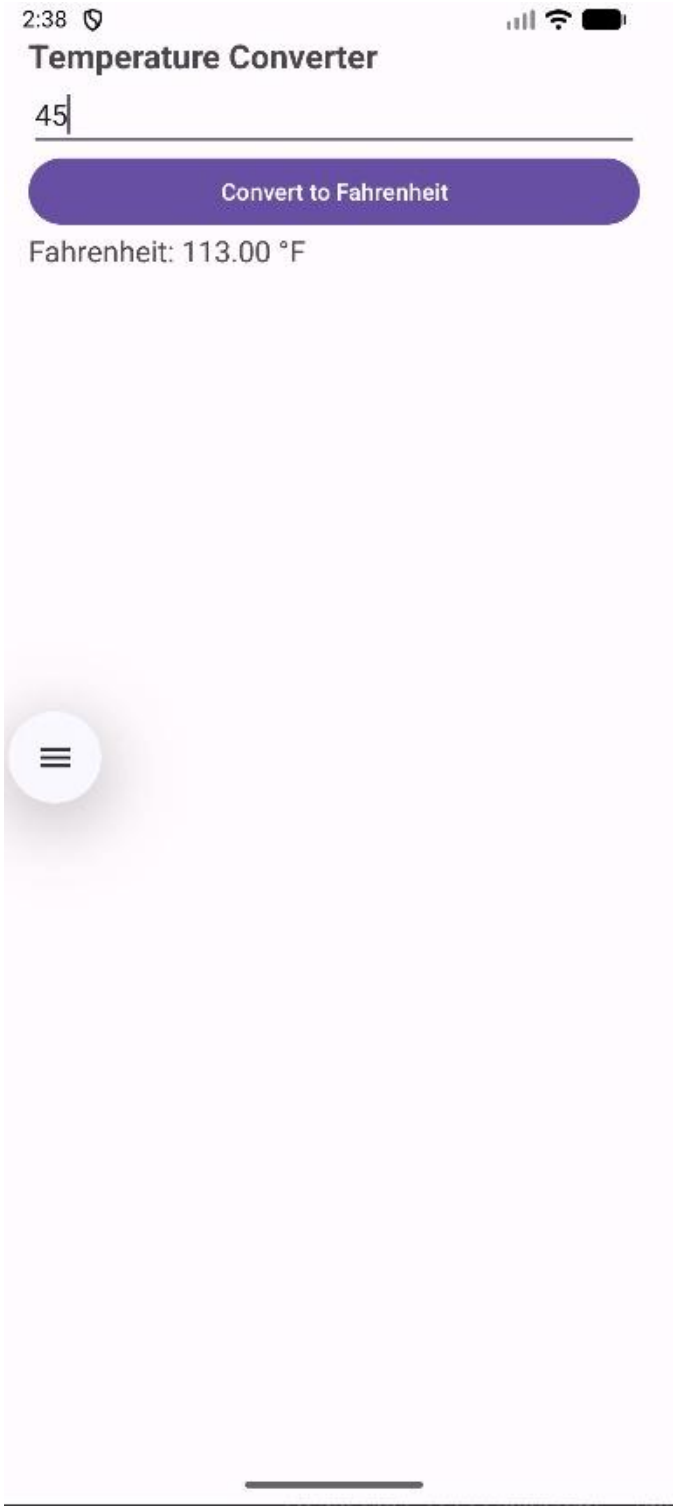
```
    android:id="@+id/tvResult"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Result will appear here"
    android:textSize="18sp" />

</LinearLayout>
```

MainActivity.java

```
package com.example.pradnyil;

import android.os.Bundle;
import android.view.View;
import android.widget.Button;
import android.widget.EditText;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

  EditText etCelsius;
  Button btnConvert;
  TextView tvResult;

  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
```

```
        etCelsius = findViewById(R.id.etCelsius);
        btnConvert = findViewById(R.id.btnConvert);
        tvResult = findViewById(R.id.tvResult);

        btnConvert.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
                if (etCelsius.getText().toString().isEmpty()) {
                    tvResult.setText("Please enter temperature");
                } else {
                    double celsius = Double.parseDouble(etCelsius.getText().toString());
                    double fahrenheit = (celsius * 9 / 5) + 32;
                    tvResult.setText("Fahrenheit: " + String.format("%.2f", fahrenheit) + " °F");
                }
            }
        });
    }
}
```

Output:


2:38
Temperature Converter
45
Convert to Fahrenheit
Fahrenheit: 113.00 °F

Q2:
What formula is used to convert temperature from Celsius to Fahrenheit?

Temperature Conversion Formula (Celsius to Fahrenheit)

$$F = \frac{9}{5}C + 32$$

- **This formula is used to convert temperature from Celsius (°C) to Fahrenheit (°F).**
- **In the given Android code, the user enters temperature in Celsius using EditText, and on button click, it is converted using this formula.**
- **The result is then displayed in TextView, showing the temperature in Fahrenheit.**

SET F

Aim: Develop a program to create a login form using the UI controls.

Q1:
Create and execute a login form with username and password fields and validate non-empty input.

Main_activity.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

  android:layout_width="match_parent"

  android:layout_height="match_parent"

  android:orientation="vertical"

  android:padding="20dp"

  android:gravity="center">


  <TextView
```

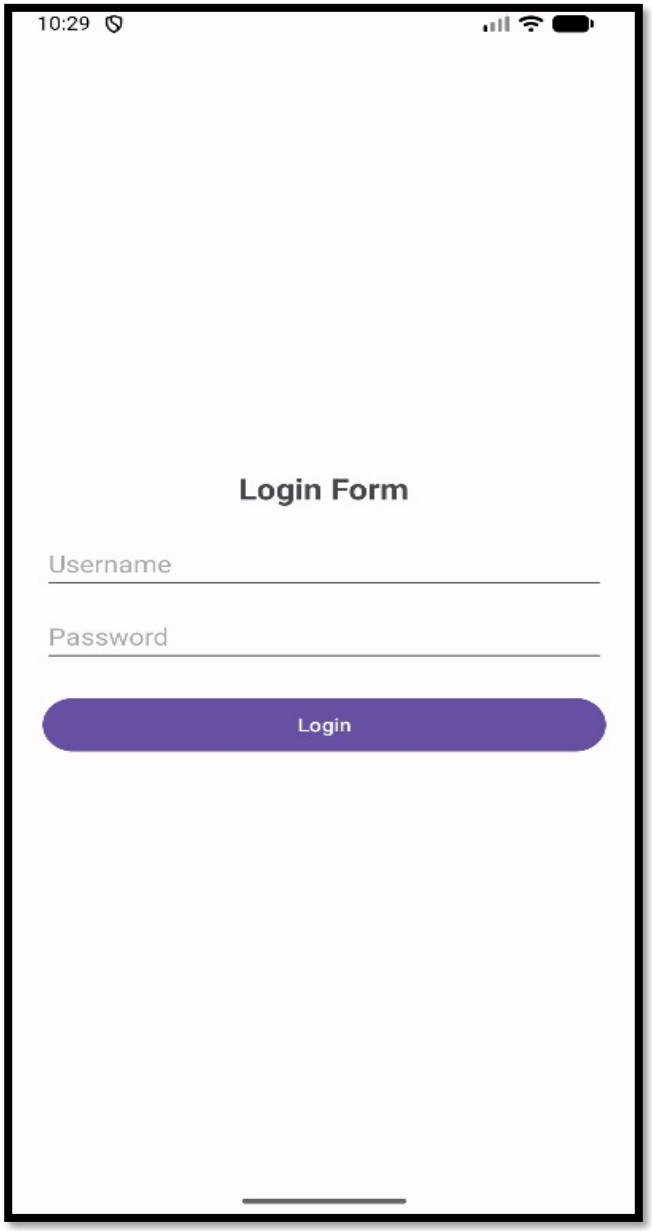
```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Login Form"
        android:textSize="22sp"
        android:textStyle="bold"
        android:layout_marginBottom="20dp" />

    <EditText
        android:id="@+id/etUsername"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Username"
        android:layout_marginBottom="10dp" />

    <EditText
        android:id="@+id/etPassword"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:hint="Password"
        android:inputType="textPassword"
        android:layout_marginBottom="20dp" />

    <Button
        android:id="@+id/btnLogin"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Login" />

</LinearLayout>
```

MainActivity.java

```
package com.example.myapplication;

import android.os.Bundle;

import android.view.View;

import android.widget.Button;

import android.widget.EditText;

import android.widget.Toast;

import androidx.appcompat.app.AppCompatActivity;


public class MainActivity extends AppCompatActivity {


  EditText etUsername, etPassword;

  Button btnLogin;


  @Override

  protected void onCreate(Bundle savedInstanceState) {

    super.onCreate(savedInstanceState);

    setContentView(R.layout.activity_main);


    etUsername = findViewById(R.id.etUsername);

    etPassword = findViewById(R.id.etPassword);

    btnLogin = findViewById(R.id.btnLogin);

      btnLogin.setOnClickListener(new View.OnClickListener() {

       @Override

       public void onClick(View v) {

          String username = etUsername.getText().toString().trim();

          String password = etPassword.getText().toString().trim();


          if (username.isEmpty() || password.isEmpty()) {
```

```
            Toast.makeText(MainActivity.this, "Please fill all fields",
Toast.LENGTH_SHORT).show();

        } else {

            Toast.makeText(MainActivity.this, "Login Successful",
Toast.LENGTH_SHORT).show();

        }

    }

  });

}
```

Output:-



Q2:
What is input validation in Android and why is it important in a login form?

- **Input validation is the process of checking whether the user input is correct, complete, and in the required format before processing it.**
- **It ensures that data entered in fields like username, email, and password is valid.**

Importance in Login Form

- **Prevents empty fields (e.g., username or password not entered).**
- **Ensures correct format (like valid email structure).**
- **Improves security by avoiding invalid or harmful input.**
- **Enhances user experience by showing proper error messages.**

SET G

Aim: Write a program to implement RadioButton.

Q1:
Write and execute code to display selected option using RadioButton.
ANS :

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

  android:layout_width="match_parent"

  android:layout_height="match_parent"

  android:orientation="vertical"

  android:padding="16dp">


  <TextView

    android:text="Select Your Hobby:"
```

```
    android:textSize="18sp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content" />

<RadioGroup
    android:id="@+id/radioGroup"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content">

    <RadioButton
        android:id="@+id/radio1"
        android:text="Reading"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" />

    <RadioButton
        android:id="@+id/radio2"
        android:text="Gaming"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" />

</RadioGroup>

<Button
    android:id="@+id/btn"
    android:text="Show"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content" />
<TextView
    android:id="@+id/result"
```

```
    android:layout_width="match_parent"
    android:layout_height="wrap_content" />

</LinearLayout>
```

MainActivity.java

```
package com.example.myapplication;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
import android.widget.RadioButton;
import android.widget.RadioGroup;
import android.widget.TextView;

public class MainActivity extends AppCompatActivity {
  @Override
  protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);

    RadioGroup radioGroup = findViewById(R.id.radioGroup);
    RadioButton radio1 = findViewById(R.id.radio1);
    RadioButton radio2 = findViewById(R.id.radio2);
    TextView result = findViewById(R.id.result);

    findViewById(R.id.btn).setOnClickListener(v -> {
      String text = "None";
      if (radio1.isChecked()) text = "Reading";
      if (radio2.isChecked()) text = "Gaming";
      result.setText("Selected: " + text);
    });
```

```
    }
}
```

OUTPUT -

6:51
Select Your Hobby:
Reading
Gaming
Show
Selected: Reading

Q2:
Explain RadioButton and its usage in Android.

- **RadioButton is a UI component that allows the user to select only one option from a group of options.**
- **It is usually used along with a RadioGroup to ensure single selection.**
- **When one RadioButton is selected, others in the same group are automatically deselected.**

Usage of RadioButton

- **Used in forms where only one choice is required (e.g., gender selection, payment method).**
- **Helps in selecting a single option from multiple choices.**
- **Improves user experience by preventing multiple selections.**

Important Attributes

- **text → Displays label of the RadioButton**
- **checked → Sets default selected option**
- **id → Unique identifier for coding**
- **onClick → Handles click events**

SET H

Aim: Design and develop a simple countdown timer.

Q1:
Create and execute a countdown timer that starts from 10 seconds and updates every second.
ANS :

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

   android:layout_width="match_parent"

   android:layout_height="match_parent"

   android:orientation="vertical"
```

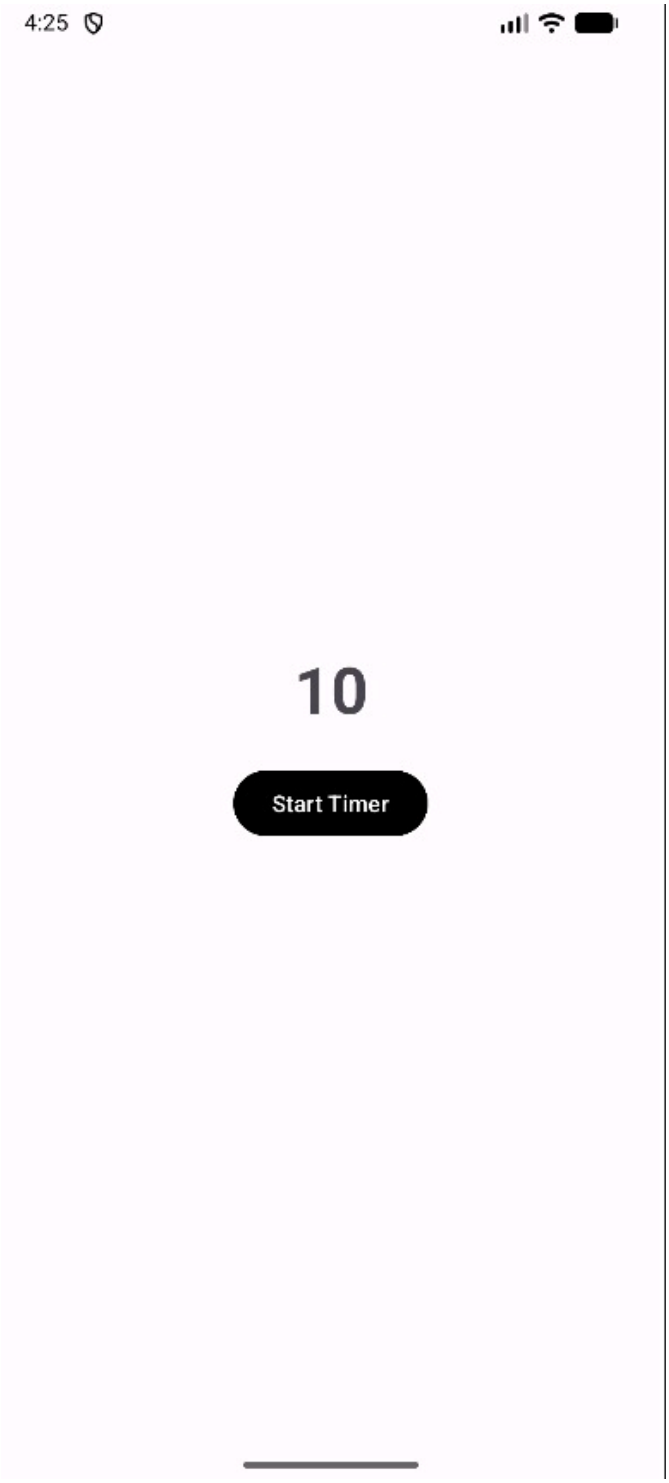
```
    android:gravity="center"
    android:padding="20dp">

    <TextView
        android:id="@+id/tvTimer"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="10"
        android:textSize="40sp"
        android:textStyle="bold"
        android:layout_marginBottom="20dp"/>

    <Button
        android:id="@+id/btnStart"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Start Timer"
        android:backgroundTint="#000000"/>
</LinearLayout>
```

MainActivity.java

```
package com.example.pradnyil;

import android.os.Bundle;
import android.os.CountDownTimer;
import android.widget.Button;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {

    TextView tvTimer;
    Button btnStart;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        tvTimer = findViewById(R.id.tvTimer);
        btnStart = findViewById(R.id.btnStart);

        btnStart.setOnClickListener(v -> {
            new CountDownTimer(10000, 1000) {
                @Override
                public void onTick(long millisUntilFinished) {
                    tvTimer.setText(String.valueOf(millisUntilFinished / 1000));
                }

                @Override
                public void onFinish() {
                    tvTimer.setText("Time Up!");
                }
            }.start();
        });
    }
}
```

Output:



Q2:
How does CountDownTimer work in Android? Explain onTick() and onFinish().

- **CountDownTimer is a class used to perform a countdown for a specified time interval.**
- **It runs in the background and updates the UI at regular intervals.**
- **In the given code, the timer runs for 10 seconds (10000 ms) and updates every 1 second (1000 ms).**

---

onTick() Method

- **onTick() is a method of CountDownTimer that is called repeatedly at fixed time intervals during the countdown.**
- **It provides the remaining time in milliseconds.**
- **In the code, it updates the TextView every second by converting milliseconds into seconds.**

---

onFinish() Method

- **onFinish() is a method of CountDownTimer that is called once when the countdown timer completes**
- **It is used to perform final actions after the countdown ends.**
- **In the code, it displays "Time Up!" in the TextView.**